

Midterm Exam: Micro 629 - Fall 2009

Texas A&M University

October 13th, 2009

“An Aggie does not lie, cheat, or steal or tolerate those who do.”

EXAM RULES

- Your interpretation of the questions makes part of the exam. Thus, no questions will be answered during the exam.
 - Choose one problem from 1 and 2. If more than one of these two problems is submitted, the one deserving the lowest score will count for your grade in the exam.
 - No personal items (bags, cellphone, books, notebooks, calculators, etc.) will be allowed with you during the exam.
1. (25/100) Let $(\mathcal{B}, C)$ be a choice structure.
 - (a) Formally state:
 - i. (3/25) WARP.
 - ii. (1/25) Sen's property α .
 - iii. (1/25) Sen's property β .
 - (b) Assume that $\mathcal{B} = 2^X \setminus \{\emptyset\}$. Prove that $(\mathcal{B}, C)$ satisfies WARP if and only if it satisfies Sen's properties α and β . Do it in two steps:
 - i. (10/25) $(\mathcal{B}, C)$ satisfies WARP $\Rightarrow$ $(\mathcal{B}, C)$ satisfies Sen's properties α and β .
 - ii. (10/25) $(\mathcal{B}, C)$ satisfies Sen's properties α and $\beta \Rightarrow (\mathcal{B}, C)$ satisfies WARP.
 2. (25/100) Let $(\mathcal{B}, C)$ be a choice structure.
 - (a) (5/25) Formally state WARP.
 - (b) Assume that $(\mathcal{B}, C)$ is a Walrasian choice structure, that is, $\mathcal{B} = \mathcal{B}^W$ and that C is single-valued. Let x be the associated demand function. Prove that $(\mathcal{B}^W, C)$ satisfies WARP if and only if x satisfies the following property: for any two price-wealth combinations, (p, w) and (p', w') , if $[p \cdot x(p', w') \leq w \text{ and } x(p', w') \neq x(p, w)]$, then $[p' \cdot x(p, w) > w']$. Do it in two steps:
 - i. (10/25) $(\mathcal{B}, C)$ satisfies WARP $\Rightarrow$ for any two price-wealth combinations, (p, w) and (p', w') , if $[p \cdot x(p', w') \leq w \text{ and } x(p', w') \neq x(p, w)]$, then $[p' \cdot x(p, w) > w']$.
 - ii. (10/25) For any two price-wealth combinations, (p, w) and (p', w') , if $[p \cdot x(p', w') \leq w \text{ and } x(p', w') \neq x(p, w)]$, then $[p' \cdot x(p, w) > w'] \Rightarrow (\mathcal{B}, C)$ satisfies WARP.
 3. (25/100) A preference $\succsim$ on $\mathbb{R}_+^L$ is convex if for each $x \in \mathbb{R}_+^L$, $U(\succsim, x)$ is convex. Assume that $\succsim$ is rational. Prove that $\succsim$ is convex if and only if for any $\{x, y\} \subseteq \mathbb{R}_+^L$ such that $x \succsim y$ and each $\alpha \in [0, 1]$, $\alpha x + (1 - \alpha)y \succsim y$. Do it in two steps:
 - (a) (12.5/25) $\succsim$ is convex $\Rightarrow$ for any $\{x, y\} \subseteq \mathbb{R}_+^L$ such that $x \succsim y$ and each $\alpha \in [0, 1]$, $\alpha x + (1 - \alpha)y \succsim y$
 - (b) (12.5/25) For any $\{x, y\} \subseteq \mathbb{R}_+^L$ such that $x \succsim y$ and each $\alpha \in [0, 1]$, $\alpha x + (1 - \alpha)y \succsim y \Rightarrow \succsim$ is convex

4. (25/100) Let $\succsim$ be a monotone and continuous rational preference on $\mathbb{R}_+^L$. Let $p \gg 0$. The **money metric representation of $\succsim$ with respect to p** is the function, denoted m_p , defined by:

$$m_p : \mathbb{R}_+^L \rightarrow \mathbb{R} \\ x \mapsto m_p(x) \equiv \min_{y \geq 0, y \succsim x} p \cdot y$$

Prove that m_p represents $\succsim$. Do it in three steps:

- (a) (5/25) Write the statement “ m_p represents $\succsim$ ” as an if and only if statement.
 - (b) (10/25) $\Rightarrow$
 - (c) (10/25) $\Leftarrow$
5. (25/100) Let $X \equiv \mathbb{R}^3$. Suppose that a Walrasian demand function x satisfies Walras’ law and is such that for each pair $(p, w) \in \mathbb{R}_{++}^L \times \mathbb{R}_+$:

$$x_1(p, w) \equiv a + b \frac{p_1}{p_3} + c \frac{p_2}{p_3},$$

$$x_2(p, w) \equiv d + e \frac{p_1}{p_3} + f \frac{p_2}{p_3}.$$

- (a) (5/25) Indicate how you would calculate $x_3(p, w)$.
 - (b) (5/25) Prove that x is homogeneous of degree zero.
 - (c) (5/25) Write $S_{lk}(p, w)$ in terms of the partial derivatives of x .
 - (d) (5/25) Calculate $S_{11}(p, w)$, $S_{12}(p, w)$, $S_{21}(p, w)$, $S_{22}(p, w)$.
 - (e) (5/25) Indicate how you would find the restrictions on the demand parameters that are implied by utility maximization. State one such a restriction.
6. (25/100 Bonus) A preference $\succsim$ on $\mathbb{R}_+^L$ is homothetic if for any $\{x, y\} \subseteq \mathbb{R}_+^L$ such that $x \sim y$ and each $\alpha \in \mathbb{R}_+$, we have that $\alpha x \sim \alpha y$. Assume that $\succsim$ is homothetic and continuous. As a first step to prove that $\succsim$ is represented by a homogeneous function of degree one, one has to prove that for any $\{x', y'\} \subseteq \mathbb{R}_+^L$ such that $x' \succsim y'$ and each $\alpha \in \mathbb{R}_+$, we have that $\alpha x' \succsim \alpha y'$. Prove this statement without assuming that the preference can be represented by a homogeneous function of degree one.

Hint: By the Intermediate Value Theorem if $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function such that for some $a \leq b$, $\varphi(a) \leq 0$ and $\varphi(b) \geq 0$, then there exists $\beta^* \in [a, b]$ such that $\varphi(\beta^*) = 0$.